

 Testleaf AI Playground for ML Machine Learning · Classification

ALGORITHM

Linear Regression

Classification

DATASET

<

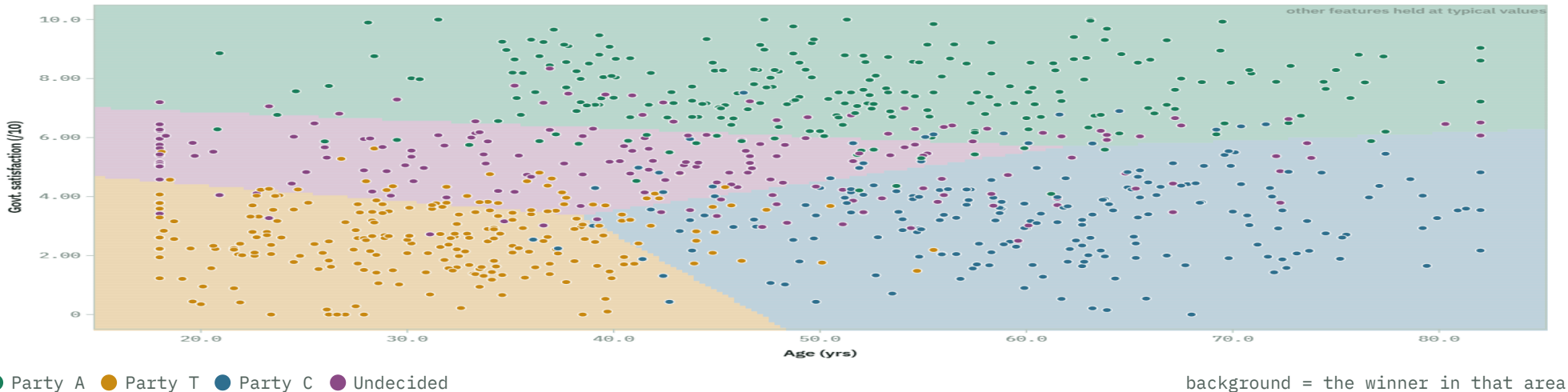
Election Preference (Multiclass)

>

trained · 672/168 split

Train again

Upload data



ROWS TRAINED ON

672

80% of the data

ROWS TESTED ON

168

held back, never seen

WHAT THIS DATA IS

Survey data with four possible answers, not two. Age and satisfaction with the current government drive the choice, with city voters leaning differently from rural ones. Parties are fictional — A, T, C — exactly as in the slide.

MAKE A PREDICTION

Age



26.0 yrs

Govt. satisfaction



0.500 /10

Urban voter

Yes

No

Yes

Predict

PREDICTED PREFERENCE

Party T

Party A

0.0%

Party T

99.9%

Party C

0.1%

Undecided

0.0%

Step 1 – each class gets its own score = base + (weight × this voter's value), added up

Party A

-9.87 pts ← base points

- 0.009 × 26.0 (Age) = -0.243

+ 1.90 × 0.500 (Govt. satisfaction) = +0.951

+ 0.476 × Yes (Urban voter) = +0.476

= -8.69 pts

Party T

12.9 pts ← base points

- 0.160 × 26.0 (Age) = -4.16

- 1.64 × 0.500 (Govt. satisfaction) = -0.818

+ 2.63 × Yes (Urban voter) = +2.63

= 10.5 pts

Party C

$2.17 \text{ pts} \leftarrow \text{base points}$
 $+ 0.067 \times 26.0 \text{ (Age)} = +1.74$
 $- 1.05 \times 0.500 \text{ (Govt. satisfaction)} = -0.524$
 $+ 0.198 \times \text{Yes (Urban voter)} = +0.198$

 $= 3.59 \text{ pts}$

Undecided

$2.69 \text{ pts} \leftarrow \text{base points}$
 $- 0.057 \times 26.0 \text{ (Age)} = -1.49$
 $+ 0.213 \times 0.500 \text{ (Govt. satisfaction)} = +0.106$
 $+ 1.15 \times \text{Yes (Urban voter)} = +1.15$

 $= 2.45 \text{ pts}$

highest score wins → Party T

Step 2a – convert each score to e^{score} (always positive)

Party A $e^{-8.69} = 0.000$
 Party T $e^{10.5} = 36,813$
 Party C $e^{3.59} = 36.1$
 Undecided $e^{2.45} = 11.6$

Total (add all 4 up) = 36,861

Step 2b – each class's share = its own e^{score} ÷ that same total (36,861)

Party A $0.000 \div 36,861 = 0.0\%$
 Party T $36,813 \div 36,861 = 99.9\%$
 Party C $36.1 \div 36,861 = 0.1\%$
 Undecided $11.6 \div 36,861 = 0.0\%$

All 4 shares add back up to 100% – that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

THE TRAINED MODEL

Party A score = **-9.87** pts $\leftarrow$ base points (starting score) - **0.009** $\times$ Age + **1.90** $\times$ Govt. satisfaction + **0.476** $\times$ Urban voter

Party T score = **12.9** pts $\leftarrow$ base points (starting score) - **0.160** $\times$ Age - **1.64** $\times$ Govt. satisfaction + **2.63** $\times$ Urban voter

Party C score = **2.17** pts $\leftarrow$ base points (starting score) + **0.067** $\times$ Age - **1.05** $\times$ Govt. satisfaction + **0.198** $\times$ Urban voter

Undecided score = **2.69** pts $\leftarrow$ base points (starting score) - **0.057** $\times$ Age + **0.213** $\times$ Govt. satisfaction + **1.15** $\times$ Urban voter

SAMPLE CALCULATION – ONE REAL ROW

45.7 yrs $\cdot$ 6.43 /10 $\cdot$ not urban voter

Party A **1.93**

Party T **-4.98**

Party C **-1.51**

Undecided **1.44**

highest wins $\rightarrow$ Party A

actual Party A

With more than two possible answers, the model keeps **one score per class** — four separate sums from the same inputs. Whichever score comes out highest is the prediction. That is all “multiclass” means.

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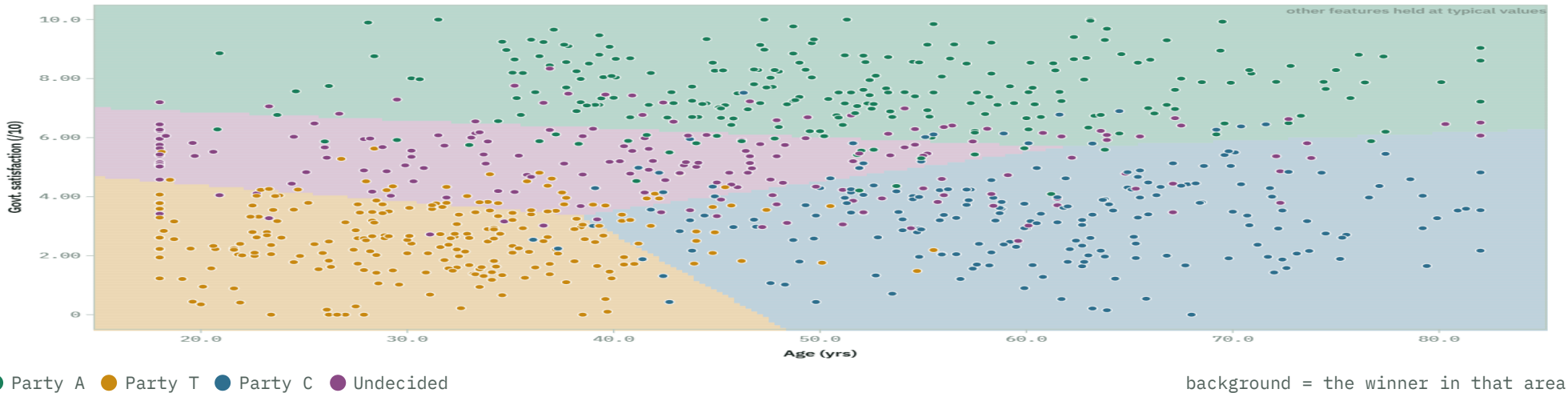
Election Preference (Multiclass)

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held back, never seen

WHAT THIS DATA IS

Survey data with four possible answers, not two. Age and satisfaction with the current government drive the choice, with city voters leaning differently from rural ones. Parties are fictional — A, T, C — exactly as in the slide.

MAKE A PREDICTION

Age



56.0 yrs

Govt. satisfaction



1.00 /10

Urban voter

Yes

No

No

Predict

PREDICTED PREFERENCE

Party C

Party A

0.0%

Party T

6.8%

Party C

92.7%

Undecided

0.5%

Step 1 – each class gets its own score = base + (weight × this voter's value), added up

Party A

-9.87 pts ← base points

- 0.009 × 56.0 (Age) = -0.524

+ 1.90 × 1.00 (Govt. satisfaction) = +1.90

+ 0.476 × No (Urban voter) = +0

= -8.49 pts

Party T

12.9 pts ← base points

- 0.160 × 56.0 (Age) = -8.97

- 1.64 × 1.00 (Govt. satisfaction) = -1.64

+ 2.63 × No (Urban voter) = +0

= 2.26 pts

Party C

$2.17 \text{ pts} \leftarrow \text{base points}$
 $+ 0.067 \times 56.0 \text{ (Age)} = +3.75$
 $- 1.05 \times 1.00 \text{ (Govt. satisfaction)} = -1.05$
 $+ 0.198 \times \text{No (Urban voter)} = +0$

 $= 4.87 \text{ pts}$

Undecided

$2.69 \text{ pts} \leftarrow \text{base points}$
 $- 0.057 \times 56.0 \text{ (Age)} = -3.20$
 $+ 0.213 \times 1.00 \text{ (Govt. satisfaction)} = +0.213$
 $+ 1.15 \times \text{No (Urban voter)} = +0$

 $= -0.306 \text{ pts}$

highest score wins → Party C

Step 2a – convert each score to e^{score} (always positive)

Party A $e^{-8.49} = 0.000$
 Party T $e^{2.26} = 9.57$
 Party C $e^{4.87} = 131$
 Undecided $e^{-0.306} = 0.736$

Total (add all 4 up) = 141

Step 2b – each class's share = its own e^{score} ÷ that same total (141)

Party A $0.000 \div 141 = 0.0\%$
 Party T $9.57 \div 141 = 6.8\%$
 Party C $131 \div 141 = 92.7\%$
 Undecided $0.736 \div 141 = 0.5\%$

All 4 shares add back up to 100% – that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

THE TRAINED MODEL

Party A score = **-9.87** pts $\leftarrow$ base points (starting score) - **0.009** $\times$ Age + **1.90** $\times$ Govt. satisfaction + **0.476** $\times$ Urban voter

Party T score = **12.9** pts $\leftarrow$ base points (starting score) - **0.160** $\times$ Age - **1.64** $\times$ Govt. satisfaction + **2.63** $\times$ Urban voter

Party C score = **2.17** pts $\leftarrow$ base points (starting score) + **0.067** $\times$ Age - **1.05** $\times$ Govt. satisfaction + **0.198** $\times$ Urban voter

Undecided score = **2.69** pts $\leftarrow$ base points (starting score) - **0.057** $\times$ Age + **0.213** $\times$ Govt. satisfaction + **1.15** $\times$ Urban voter

SAMPLE CALCULATION – ONE REAL ROW

45.7 yrs $\cdot$ 6.43 /10 $\cdot$ not urban voter

Party A **1.93**

Party T **-4.98**

Party C **-1.51**

Undecided **1.44**

highest wins $\rightarrow$ Party A

actual Party A

With more than two possible answers, the model keeps **one score per class** — four separate sums from the same inputs. Whichever score comes out highest is the prediction. That is all “multiclass” means.



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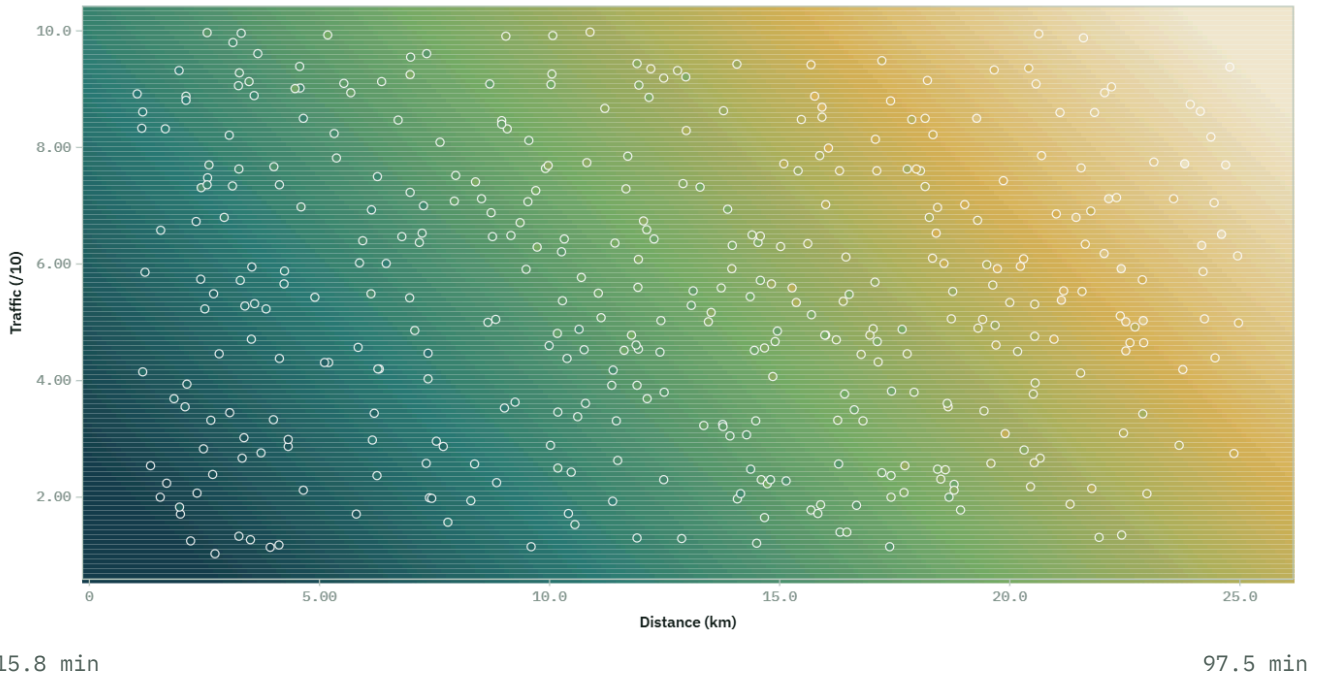
Food Delivery Time

>

trained · 320/80 split

Train again

Upload data



ROWS TRAINED ON

320

80% of the data

ROWS TESTED ON

80

held back, never seen

WHAT THIS DATA IS

Distance and traffic both push the delivery time up at a steady rate.

MAKE A PREDICTION

Distance



5.50 km

Traffic



9.10 /10

Predict

TOTAL TIME

49.5 min

Base time – the starting value, before any feature counts

7.56 min

Distance

2.40 min per km × 5.50 km

= +13.2 min

Traffic

$$\begin{aligned} & 3.16 \text{ min per point} \times 9.10 / 10 \\ & = +28.7 \text{ min} \end{aligned}$$

TOTAL 49.5 min

THE TRAINED MODEL

Delivery_Minutes in min =

$$\begin{aligned} & 7.56 \text{ min} \leftarrow \text{base time (minimum)} \\ & + 2.40 \text{ min per km} \times \text{Distance} \\ & + 3.16 \text{ min per point} \times \text{Traffic} \end{aligned}$$

SAMPLE CALCULATION — ONE REAL ROW FROM THE DATA

$$8.96 \text{ km} \cdot 8.46 / 10$$

$$\begin{aligned} & 7.56 \text{ (base)} \\ & + 2.40 \times 8.96 = +21.5 \\ & + 3.16 \times 8.46 = +26.7 \end{aligned}$$

model says 55.8 min

actual value 55.4 min

Read it as a rate card. The first number is the **base time** — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.



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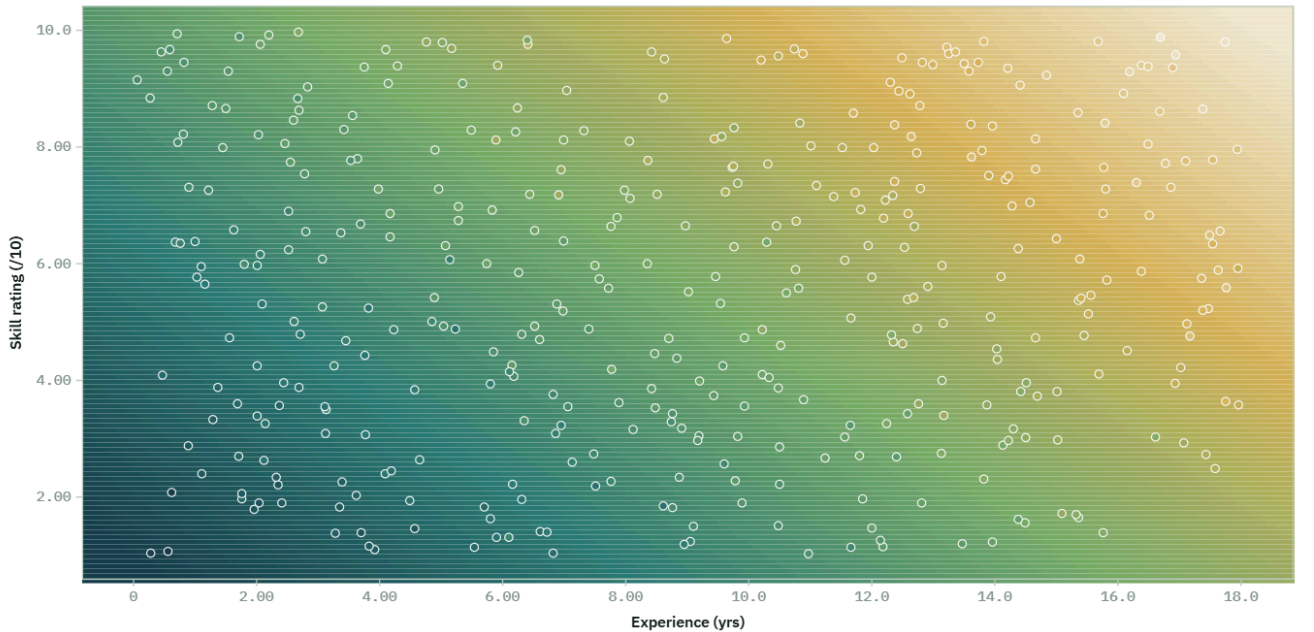
QA Engineer Salary

>

trained · 320/80 split

Train again

Upload data



4.37 LPA

34.2 LPA

ROWS TRAINED ON

320

80% of the data

ROWS TESTED ON

80

held back, never seen

WHAT THIS DATA IS

Experience and skill rating each add a predictable amount to the package.

MAKE A PREDICTION

Experience



15.5 yrs

Skill rating



8.50 /10

Predict

TOTAL PACKAGE

29.5 LPA

Base salary – the starting value, before any feature counts

3.10 LPA

Experience

0.966 LPA per year × 15.5 yrs

= +15.0 LPA

Skill rating

$$\begin{aligned} & 1.34 \text{ LPA per point} \times 8.50 / 10 \\ & = +11.4 \text{ LPA} \end{aligned}$$

TOTAL 29.5 LPA

THE TRAINED MODEL

Salary_LPA in LPA =

$$\begin{aligned} & 3.10 \text{ LPA} \leftarrow \text{base salary (minimum)} \\ & + 0.966 \text{ LPA per year} \times \text{Experience} \\ & + 1.34 \text{ LPA per point} \times \text{Skill rating} \end{aligned}$$

SAMPLE CALCULATION — ONE REAL ROW FROM THE DATA

9.46 yrs · 5.78 /10

$$\begin{aligned} & 3.10 \text{ (base)} \\ & + 0.966 \times 9.46 = +9.14 \\ & + 1.34 \times 5.78 = +7.75 \end{aligned}$$

model says 20.0 LPA

actual value 19.4 LPA

Read it as a rate card. The first number is the **base salary** — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.